Research Proposal

Generative AI and RAG-based Framework for Automated Football Match Summaries and Tactical Insights

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Abstract

The rapid advancement of artificial intelligence has significantly transformed sports analytics, particularly in the domain of football match analysis and automated content generation. Traditional approaches for generating match summaries and commentary primarily rely on manual processes or basic text-based techniques, which often fail to capture the rich contextual information embedded in structured match data.

Recent research has explored multimodal frameworks that integrate video, audio, and textual data to produce more comprehensive and engaging outputs. However, such approaches are computationally expensive, complex to implement, and difficult to scale in resource-constrained environments.

In addition, existing football analytics systems often focus on isolated tasks such as event detection or commentary generation, lacking the ability to perform holistic reasoning and contextual understanding across multiple aspects of the game. This limitation restricts their capability to generate meaningful tactical insights and coherent narratives.

To address these challenges, this research proposes a Generative AI and Retrieval-Augmented Generation (RAG)-based framework for automated football match summaries and tactical insights. The proposed system adopts a cost-efficient and scalable approach by leveraging structured football event data, rather than relying on multimodal inputs. It integrates large language models with retrieval mechanisms to enhance contextual awareness, improve factual consistency, and reduce hallucinations. Furthermore, the framework incorporates agentic AI concepts to enable task decomposition and improved reasoning for complex football analysis tasks.

The proposed methodology involves transforming structured event data into textual representations, embedding and indexing them in a vector database, and generating responses through a multi-stage agentic pipeline. The system is evaluated using both quantitative metrics and qualitative assessment to measure the relevance, coherence, and accuracy of generated outputs.

This research aims to develop an intelligent, scalable, and resource-efficient AI system that enhances automated football analytics, while demonstrating the effectiveness of structured-data-driven RAG approaches as a practical alternative to multimodal systems.

Table of Contents

Topics

Abstract i
List of Figures 1
List of Tables 1
List of Abbreviations 1
1. Background 2
2. Literature Review 2
2.1. Computer Vision in Sports Analytics 2
2.2. Multimodal AI for Sports Understanding 3
2.3. Conversational and Interactive Systems in Sports 3
2.4. Agentic AI for Comprehensive Sports Understanding 3
2.5. Comparative Analysis of Existing Approaches 4
2.6. Research Gap 5
3. Problem Statement 6
4. Research Questions 6
5. Aim and Objectives 7
5.1. Aim 7
5.2. Objectives 7
5.3. Research Contributions 7
6. Research Methodology 8
6.1. Research design 8
6.2. System overview 9
6.2.1. System Architecture Overview 9
6.2.2. Key Components of the System 11
6.2.3. Agent-Based Design 11
6.2.4. Key Features of the Proposed System 11
6.2.5. Advantages over Traditional Approaches 12
6.3. Agent Design 12
6.3.1. Query Understanding Agent 12
6.3.2. Retriever Agent 12
6.3.3. Context Aggregator Agent 12
6.3.4. Reasoning Agent 13
6.3.5. Generator Agent 13
6.3.6. Evaluator Agent 13
6.3.7. Design Rationale 13

Table of Contents

6.4. Data sources and Transformation 13
6.4.1. Structured Event Data 13
6.4.2. Textual Data 14
6.4.3. Metadata 14
6.4.4. Data Preprocessing 14
6.4.5. Justification of Data Selection 14
6.5. Data ingestion and metadata schema 15
6.6. Chunking strategy 15
6.7. Embeddings and Vector DB setup 16
6.8. Retrieval Strategy 17
6.9. Retrieval Metrics 17
6.10. Response Generation and Prompting 18
6.11. Evaluation Plan 18
6.11.1. Evaluation Setup 18
6.11.2. Retrieval Evaluation 19
6.11.3. Generation Evaluation 19
6.11.4. Human Evaluation 19
6.11.5. Baseline Comparison 20
6.11.6. Ablation Study 20
6.11.7. Success Criteria 20
6.11.8. Error Analysis 21
6.12. Implementation Details 21
6.13. Budget and Resource Considerations 21
6.14. Ethical Considerations 22
7. Requirements and Resources 22
7.1. Data Requirements 22
7.2. Software Requirements 23
7.3. Hardware Requirements 23
7.4. Computational Resources and Budget 23
7.5. Resource Constraints and Design Considerations 24
8. Project Timeline and Gantt Chart 24
8.1. Project Timeline 24
8.2. Gantt Chart Representation 25
8.3. Timeline Justification 26
References 27

List of Figures

Figure Number	Description	Page Number
Figure 1	Agentic RAG System Architecture for Football Match Analysis	10
Figure 2	Project Timeline	25
Figure 3	Gantt Chart	26

List of Tables

Table Number	Description	Page Number
Table 1	Comparative Analysis of Existing Approaches	4
Table 2	Key Components	11
Table 3	Agents and Roles	11
Table 4	Event Data Schema	15
Table 5	Project Timeline	24

List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BM25	Best Matching 25
BLEU	Bilingual Evaluation Understudy
ChromaDB	Chroma Database
CPU	Central Processing Unit
DSR	Design Science Research
FAISS	Facebook AI Similarity Search
GPU	Graphics Processing Unit
JSON	JavaScript Object Notation
LLM	Large Language Model
NLP	Natural Language Processing
P@K	Precision at K
R@K	Recall at K
RAG	Retrieval-Augmented Generation
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
SBERT	Sentence-BERT
VS Code	Visual Studio Code

1. Background

The field of sports analytics, particularly in football, has undergone significant transformation with the advancement of artificial intelligence and data-driven technologies. Traditional approaches to football analysis and match summarisation have largely relied on manual commentary, statistical reports, or rule-based Natural Language Processing (NLP) systems, which are limited in capturing complex contextual relationships within match events.

Recent developments in artificial intelligence have introduced more advanced methods for automated sports analysis and narrative generation. In particular, the increasing availability of structured football event data has enabled data-driven approaches to represent match events in a machine-readable form and support automated understanding of game dynamics (Giancola et al., 2018). At the same time, multimodal approaches combining video, audio, and textual information have demonstrated the potential to improve contextual richness in football analytics and summarisation tasks (Sarkhoosh et al., 2024).

In parallel, Generative AI and Large Language Models (LLMs) have created new possibilities for producing coherent and human-like narratives from domain-specific information. Retrieval-Augmented Generation (RAG) further strengthens these systems by enabling language models to access external knowledge sources, thereby improving factual grounding and reducing hallucinations (Lewis et al., 2020).

Overall, the current landscape of football analytics reflects a transition from manually driven and isolated analytical methods toward intelligent, data-driven, and context-aware systems. These developments provide a strong foundation for further research into intelligent and efficient methods for automated football analysis, summarisation, and insight generation.

2. Literature Review

Recent advancements in Artificial Intelligence (AI) have significantly transformed the field of sports analytics, particularly in football, by enabling automated understanding, analysis, and enhanced viewer engagement. This section reviews key research contributions across computer vision, multimodal AI, conversational systems, and agentic frameworks.

2.1. Computer Vision in Sports Analytics

Computer vision has played a foundational role in analysing sports videos by enabling tasks such as object detection, player tracking, and event recognition. The work on ASTRA (Action Spotting Transformer) introduces a transformer-based architecture for detecting and localizing key events in soccer videos (Xarles et al., 2023). The model addresses challenges such as long-tail data distribution, temporal ambiguity, and noisy annotations by integrating encoder-decoder architectures and uncertainty-aware prediction mechanisms.

Such approaches demonstrate the effectiveness of deep learning models in extracting structured insights from unstructured video data. However, these systems are primarily limited to low-level perception tasks such as event detection and lack higher-level contextual understanding. They do not capture the broader tactical context or generate meaningful narratives, thereby restricting their applicability in real-world analytical and decision-making scenarios.

2.2. Multimodal AI for Sports Understanding

To overcome the limitations of unimodal systems, recent research has focused on integrating multiple data modalities, including video, audio, and text. The AiCommentator system presents a multimodal conversational agent that combines computer vision and natural language processing to generate automated and interactive football commentary (Andrews et al., 2024). By incorporating embedded visualizations and user interaction mechanisms, the system enhances viewer engagement and understanding of in-game events.

Similarly, SoccerSum introduces a multimodal framework that integrates video analysis, audio processing, and large language models to generate automated summaries and storytelling for soccer events (Sarkhoosh et al., 2024). The system leverages object detection, audio intensity analysis, and metadata extraction to produce context-rich narratives tailored for social media platforms.

The study *Generating Live Soccer-Match Commentary from Play Data* (Taniguchi et al., 2019) explores the generation of live soccer commentary from structured play-by-play event data, demonstrating the feasibility of transforming event-level representations into natural language narratives. However, the approach remains primarily limited to direct data-to-text generation and does not incorporate retrieval-based grounding or advanced reasoning mechanisms.

While these approaches demonstrate the potential of multimodal AI in enriching sports analytics, they are often built on predefined pipelines and lack adaptive intelligence. The integration of multiple modalities does not necessarily guarantee deeper understanding, as these systems struggle to align heterogeneous data sources effectively. Additionally, they lack mechanisms for dynamic reasoning and contextual retrieval, limiting their ability to handle complex analytical queries.

2.3. Conversational and Interactive Systems in Sports

Conversational AI has emerged as a promising approach to enhance user interaction in sports analytics. Systems like AiCommentator demonstrate how conversational agents can simulate human-like commentary and provide real-time insights through natural language interfaces (Andrews et al., 2024). These systems enable users to query information, interact with match data, and receive contextual explanations.

Despite these advancements, most conversational systems operate within predefined templates or limited interaction scopes. They often lack deeper reasoning capabilities and are unable to dynamically adapt to evolving match contexts or complex user queries. Furthermore, these systems typically depend on static knowledge sources, which restricts their ability to provide up-to-date and contextually relevant information.

2.4. Agentic AI for Comprehensive Sports Understanding

Recent developments in agentic AI have introduced multi-agent systems capable of collaborative reasoning and task decomposition. The SoccerAgent framework proposes a multi-agent architecture that integrates multiple tools and knowledge sources to achieve comprehensive soccer understanding (Rao et al., 2025). By leveraging a domain-specific knowledge base (SoccerWiki) and benchmark datasets (SoccerBench), the system can handle diverse tasks such as question answering, event analysis, and commentary generation.

This approach represents a shift from isolated models toward more holistic and scalable systems capable of reasoning across multiple tasks and data sources. However, such systems are still in early stages of development and face challenges related to real-time processing, scalability, and integration with live multimodal data streams. Additionally, the complexity of coordinating multiple agents introduces overhead and may impact system efficiency in practical deployments.

2.5. Comparative Analysis of Existing Approaches

To better understand the current landscape of artificial intelligence in football analytics, a comparative evaluation of existing approaches is presented in Table 2.1.

Table 1: Comparative Analysis of Existing Approaches

Approach	Strength	Limitation
Computer Vision Models	Accurate event detection and player tracking	Lack higher-level contextual and tactical understanding
Multimodal AI Systems	Rich integration of video, audio, and text data	Weak reasoning capabilities and challenges in modality alignment
Conversational Systems	Enable interactive and user-friendly insights	Limited intelligence; often rely on predefined templates
Agentic AI Systems	Strong reasoning and task decomposition capabilities	High computational complexity and scalability challenges
RAG-based Systems	Context-aware retrieval and dynamic query handling	Limited reasoning and weak integration with multi-modal data

Analytical Discussion

The comparative analysis highlights that existing approaches address different dimensions of football analytics but remain fundamentally fragmented.

Computer vision models are highly effective in extracting structured information from unstructured video data through tasks such as event detection and player tracking. However, these models are primarily limited to perception-level understanding and do not capture higher-level tactical insights or contextual relationships within the game.

Multimodal AI systems attempt to enhance this understanding by integrating multiple data sources, including video, audio, and text. While this leads to richer and more expressive outputs, these systems face challenges in aligning heterogeneous modalities and often lack the ability to perform deeper reasoning or adapt dynamically to complex analytical queries.

Conversational systems improve accessibility and user interaction by enabling natural language interfaces for querying match-related information. Despite this, they are generally constrained by predefined logic or templates and lack the intelligence required for advanced analytical reasoning.

Agentic AI frameworks represent a significant advancement by introducing multi-agent architectures capable of collaborative reasoning and task decomposition. These systems demonstrate strong analytical capabilities; however, their complexity and computational overhead limit their scalability and practical deployment in real-time scenarios.

Retrieval-Augmented Generation (RAG) approaches further extend football analytics by grounding generation in external knowledge sources. In particular, conventional RAG improves response relevance through semantic retrieval over embedded text or event-derived chunks, but it remains limited when queries require richer relational or multi-step reasoning. Knowledge graph RAG improves this by explicitly modelling entities and relationships, which is valuable in football settings involving players, teams, events, and match situations, although it introduces additional modelling complexity (Sepasdar, 2024). A more balanced approach is hybrid RAG, which combines semantic retrieval with structured data, metadata, and contextual filtering to provide stronger contextual grounding while remaining more practical than fully graph-dependent approaches. This makes hybrid RAG especially relevant for football analytics tasks requiring both accuracy and analytical flexibility.

Overall, the analysis reveals that current approaches focus on isolated capabilities—such as perception, data integration, interaction, retrieval, or reasoning—without effectively combining them into a unified framework. This fragmentation underscores the need for an integrated solution that can simultaneously leverage retrieval mechanisms, multi-modal understanding, and reasoning capabilities to generate accurate, context-aware, and actionable football insights.

2.6. Research Gap

Despite significant advancements in football analytics and automated commentary generation using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), several critical limitations persist in existing approaches.

Current RAG-based football analytics systems primarily focus on structured event data but lack the ability to dynamically adapt to complex, context-rich queries. These systems often fail to incorporate deeper contextual factors such as opponent strength, match importance, and temporal dynamics, resulting in limited interpretability and actionable insights.

In parallel, recent multimodal approaches integrating video, audio, and textual data have demonstrated improvements in commentary generation. However, they introduce significant computational complexity and face challenges in aligning heterogeneous data modalities while accurately capturing fine-grained interactions between players, events, and game context.

Furthermore, many existing systems rely on single-pass generation or task-specific models, which restrict their ability to perform multi-step reasoning. Even advanced frameworks tend to address isolated tasks and lack a unified mechanism for holistic understanding across multiple dimensions of football analytics.

Therefore, there remains no lightweight, structured-data-driven framework that effectively combines computational efficiency, retrieval-based grounding, and agentic reasoning for football match summarisation and tactical insight generation. This research addresses that gap through a lightweight Agentic RAG framework for tactical football analysis.

3. Problem Statement

Football analytics systems increasingly rely on large volumes of structured event data, player statistics, and contextual match information. Despite the growing availability of such data, existing systems still struggle to transform it into coherent, context-aware, and analytically meaningful narratives that can support match understanding and tactical interpretation.

A central limitation is that many current approaches either depend on computationally intensive multimodal pipelines or remain restricted to shallow descriptive outputs. Multimodal systems can provide rich contextual representations, but they are difficult to scale in resource-constrained environments and often face challenges in aligning heterogeneous data sources effectively (Sarkhoosh et al., 2024; Li et al., 2025). On the other hand, conventional data-to-text and RAG-based approaches improve automation and factual grounding, yet they frequently lack the reasoning capability needed to generate deeper tactical insights from structured football data (Taniguchi et al., 2019; Tzikas et al., 2025).

As a result, current football analytics solutions often produce outputs that are fragmented, overly descriptive, or insufficiently adaptable to complex user queries. Even more advanced systems that introduce collaborative reasoning remain difficult to apply in lightweight and scalable settings due to architectural complexity (Rao et al., 2025).

Therefore, the core research problem addressed in this study is the absence of a lightweight, structured-data-driven Agentic RAG framework capable of transforming football event data into context-aware match summaries and tactical insights through retrieval-based grounding and multi-step reasoning.

Problem Statement

Current football analytics systems are limited in their ability to transform structured football event data into grounded, coherent, and analytically meaningful narratives. Existing approaches either rely on computationally expensive multimodal processing or lack the reasoning depth required for tactical interpretation (Li et al., 2025; Tzikas et al., 2025). There is therefore a need for a lightweight Agentic RAG framework that can generate accurate, context-aware football match summaries and tactical insights in a scalable and resource-efficient manner.

4. Research Questions

This study aims to investigate the effectiveness of integrating Agentic Retrieval-Augmented Generation (Agentic RAG) for automated football match summarization and tactical insight generation. The following research questions guide this research:

- **RQ1:** How does an Agentic RAG-based framework improve contextual accuracy and adaptability in football match summarization compared to a naive RAG approach?
- **RQ2:** What is the impact of combining structured event representations, textual contextual sources, and metadata-aware retrieval on the quality and relevance of generated football insights?

- **RQ3:** How does agent-based reasoning influence the quality of tactical interpretation compared to single-pass LLM-based generation?
- **RQ4:** What evaluation metrics and methodologies are most effective in assessing the performance of Agentic RAG systems in sports analytics tasks?

5. Aim and Objectives

5.1. Aim

The aim of this research is to design and develop a **lightweight Agentic Retrieval-Augmented Generation (Agentic RAG) framework** for automated football match summarisation and tactical insight generation. The proposed system leverages structured football event data to produce accurate, context-aware, and coherent natural language narratives, while improving factual consistency and reducing hallucinations in generated outputs.

In contrast to conventional RAG-based approaches, this research incorporates agent-based reasoning to enable task decomposition, multi-step analysis, and enhanced interpretability, thereby improving the system's ability to generate meaningful tactical insights in a resource-efficient manner.

5.2. Objectives

To achieve the above aim, this research is guided by the following objectives:

1. To analyse existing approaches in football analytics, automated summarisation, Retrieval-Augmented Generation (RAG), and agentic AI in order to identify key limitations and research gaps.
2. To design a structured data processing pipeline that transforms football event data into textual representations suitable for semantic retrieval and downstream language generation.
3. To develop a lightweight Agentic RAG architecture that integrates retrieval, metadata-aware filtering, and agent-based reasoning for football match summarisation and tactical insight generation.
4. To implement an effective retrieval strategy incorporating semantic search, chunking, and metadata-aware retrieval to improve the relevance and quality of retrieved evidence.
5. To generate coherent football match summaries and tactical insights grounded in retrieved context using a large language model.
6. To evaluate the proposed framework against a naïve RAG baseline using retrieval metrics, generation quality metrics, and human evaluation criteria such as groundedness, coherence, and tactical usefulness.

5.3. Research Contributions

This research aims to advance the application of Generative AI in football analytics by introducing a novel, efficient, and domain-adapted framework. While prior work has explored Retrieval-Augmented Generation (RAG) and multi-agent systems in sports analytics, significant gaps remain in terms of efficiency, adaptability to structured data, and depth of tactical reasoning (Tzikas et al., 2025).

To address these limitations, this study makes the following key contributions:

1. **Lightweight Agentic RAG Framework for Structured Football Data** Unlike existing multimodal or computationally intensive approaches that rely on video and audio processing (Li et al., 2025), this research proposes a lightweight agentic RAG architecture designed specifically for structured football event data. The system employs specialised agents (retriever, reasoning, and generator) to enable multi-step analytical reasoning while maintaining computational efficiency and scalability.
2. **Hierarchical Chunking with Metadata-Aware Retrieval** The study introduces a novel data representation strategy that combines hierarchical chunking of structured football events with metadata-aware retrieval. This approach preserves temporal, spatial, and contextual relationships within match data, improving retrieval relevance and enabling more accurate generation of tactical insights compared to flat chunking methods used in existing RAG systems.
3. **Comparative Evaluation Against Naive RAG for Tactical Insight Generation** In contrast to prior research that focuses primarily on descriptive analytics or commentary generation (Tzikas et al., 2025), this work conducts a systematic comparative evaluation between the proposed agentic RAG framework and a baseline naïve RAG system. The evaluation specifically targets the generation of tactical insights, using both retrieval and generation quality metrics, thereby establishing a benchmark for effectiveness in football analytics applications.

6. Research Methodology

6.1. Research design

This study adopts a **Design Science Research (DSR)** methodology, as the primary objective is to develop a novel computational artefact—an **Agentic Retrieval-Augmented Generation (Agentic RAG)** system for football analysis. DSR is particularly suitable for this research, as it facilitates knowledge creation through the design, development, and evaluation of artefacts that address real-world problems (Peffers et al., 2007).

The research follows key stages of the DSR process:

Problem Definition:

Identifying limitations in existing football analytics systems, particularly the lack of efficient, context-aware, and grounded analysis without reliance on computationally expensive multimodal inputs.

Objective Specification:

Defining the goal of developing an accurate, scalable, and resource-efficient system for automated match summarisation and tactical insight generation.

Design and Development:

Constructing an Agentic RAG-based architecture that integrates retrieval mechanisms with agent-based reasoning to enhance contextual understanding and analytical capability.

Evaluation:

Assessing system performance through comparison with a naïve RAG baseline, focusing on both retrieval effectiveness and the quality of generated outputs.

The research follows an **iterative build-and-evaluate approach**, ensuring that design decisions—such as the adoption of an agent-based architecture—are empirically validated rather than assumed.

To complement the DSR paradigm, an **experimental methodology** is employed to quantitatively evaluate system performance. The proposed Agentic RAG system is benchmarked against a naïve RAG baseline across key dimensions, including retrieval relevance and generation quality. This aligns with established evaluation practices in retrieval-augmented systems(Lewis et al., 2020).

Furthermore, the study incorporates **explicit resource constraints** as part of the research design. Due to limited computational resources, the system excludes video and audio processing and instead relies on publicly available structured datasets. This constraint-driven approach ensures feasibility, scalability, and reproducibility, while reflecting practical considerations in real-world deployment of Generative AI systems.

While this research is framed as a "Generative AI and RAG-based framework," the proposed solution specifically adopts an Agentic RAG architecture, extending traditional RAG by incorporating agent-based reasoning and modular processing for improved contextual understanding and analytical capability.

6.2. System overview

The proposed system accepts a user's textual query (e.g., "Summarize the match and provide key tactical insights") and generates a structured response comprising a narrative summary, key moments, and tactical observations. The system is based on an Agentic Retrieval-Augmented Generation (Agentic RAG) framework, where a large language model operates over retrieved, contextually relevant information to ensure grounded and reliable outputs (Lewis et al., 2020).

Unlike naive RAG systems, which rely on a single-pass retrieval and generation process, the proposed framework introduces a **modular and multi-stage pipeline** that enables deeper contextual understanding and structured reasoning.

6.2.1. System Architecture Overview

The overall workflow of the proposed system is illustrated as a sequential pipeline:

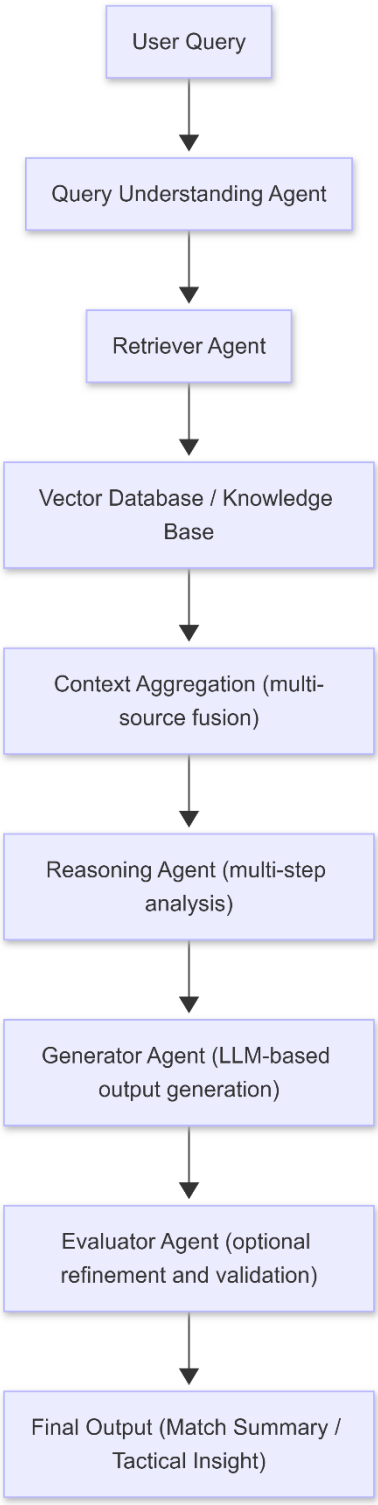


Figure 1: Agentic RAG System Architecture for Football Match Analysis

The figure demonstrates the sequential interaction between query processing, retrieval mechanisms, and response generation. This modular architecture ensures that each stage of the

pipeline is transparent, controllable, and optimisable, thereby improving both retrieval accuracy and output quality.

6.2.2. Key Components of the System

The system is composed of the following key components:

Table 2: Key Components

Component	Description
User Query Interface	Accepts natural language queries related to football matches
Query Understanding Module	Extracts intent, entities (teams, players), and context from the query
Retrieval Module	Fetches relevant information from structured datasets and vector databases
Context Aggregation Module	Combines retrieved information from multiple sources into a unified context
Reasoning Module	Performs multi-step analysis to derive tactical insights
Generation Module	Produces natural language summaries using a Large Language Model
Evaluation Module	Validates and refines outputs to improve coherence and reduce hallucinations

6.2.3. Agent-Based Design

The proposed framework introduces multiple agents, each responsible for a specific function within the pipeline:

Table 3: Agents and Roles

Agent	Role
Query Understanding Agent	Interpretation
Retriever Agent	Data Access
Context Aggregator Agent	Integration
Reasoning Agent	Analysis
Generator Agent	Output Generation
Evaluator Agent (optional)	Validation

This modular design enables **task specialization**, improves interpretability, and allows flexible extension of the system.

6.2.4. Key Features of the Proposed System

The proposed Agentic RAG framework introduces the following key features:

Context-Aware Retrieval: Uses hybrid retrieval (semantic + keyword + metadata filtering) to improve relevance.

Agent-Based Reasoning: Enables multi-step analytical processing instead of single-pass generation.

Resource-Efficient Design: Avoids direct video/audio processing by relying on structured and textual data

Modular Architecture: Allows independent optimization of retrieval, reasoning, and generation components.

6.2.5. Advantages over Traditional Approaches

Compared to naive RAG and single-pass LLM systems, the proposed framework:

- Improves **contextual accuracy** through enhanced retrieval
- Enables **deeper reasoning** via agent-based architecture Produces more **interpretable and structured outputs**
- Maintains **scalability** by avoiding computationally expensive multi-modal processing

6.3. Agent Design

The proposed Agentic RAG framework adopts a **multi-agent architecture**, where each agent is responsible for a specific task within the overall pipeline. This design promotes modularity, task specialization, and improved interpretability compared to traditional monolithic systems. Each agent operates independently while sharing contextual information, enabling structured and multi-step reasoning for football analytics.

6.3.1. Query Understanding Agent

The Query Understanding Agent is responsible for interpreting the user's input and preparing it for downstream processing. It identifies the intent of the query—such as whether the user is requesting a match summary or tactical insight—and extracts key entities including teams, players, events, and timeframes. The extracted information is then normalized into a structured format to ensure accurate and efficient retrieval. This step is essential to ensure that subsequent components operate on a well-defined query representation.

6.3.2. Retriever Agent

The Retriever Agent is responsible for fetching relevant information from the knowledge base using a hybrid retrieval strategy. It combines semantic search through vector embeddings with keyword-based methods such as BM25, while also applying metadata filters (e.g., team names, match identifiers, event types) to improve precision. The retrieved data forms the contextual foundation for subsequent reasoning and generation stages.

6.3.3. Context Aggregator Agent

The Context Aggregator Agent integrates information obtained from multiple sources into a unified and structured representation. It combines structured event data, textual descriptions, and metadata, while filtering out redundant or irrelevant content. The aggregated context is organized in a format suitable for analytical processing, ensuring that the system operates on coherent and comprehensive input.

6.3.4. Reasoning Agent

The Reasoning Agent serves as the core analytical component of the system. It performs multi-step reasoning over the aggregated context to identify patterns and derive tactical insights. This includes analysing event sequences, identifying trends such as attacking dominance or possession shifts, and performing logical inference to generate meaningful interpretations. Unlike naive RAG systems, which rely on direct generation from retrieved context, this component enables deeper analytical understanding.

6.3.5. Generator Agent

The Generator Agent is responsible for producing the final output in natural language using a Large Language Model (LLM). It transforms the structured insights generated by previous components into coherent and readable summaries or tactical explanations. The agent ensures fluency and contextual relevance by incorporating both retrieved information and reasoning outputs into the generated response.

6.3.6. Evaluator Agent

The Evaluator Agent is responsible for validating and refining the generated output. It assesses the relevance, coherence, and factual consistency of the response, while also detecting potential hallucinations or inconsistencies. When necessary, it triggers refinement steps to improve output quality, thereby enhancing the overall reliability of the system.

6.3.7. Design Rationale

The adoption of a multi-agent architecture addresses key limitations of traditional RAG systems, which typically rely on a single retrieval and generation step without explicit reasoning. By decomposing the process into specialized components, the proposed framework enables improved reasoning capability, better interpretability of outputs, and greater flexibility for system extension. Additionally, the modular design supports scalability and allows individual components to be optimized independently.

6.4. Data sources and Transformation

The proposed system relies on **structured and textual data sources** to support retrieval, reasoning, and generation tasks. In line with the resource constraints defined in the research design, the study avoids direct processing of computationally intensive video and audio data, and instead utilises **publicly available football datasets and derived textual information**.

6.4.1. Structured Event Data

Structured event data forms the primary source of information for the system. This includes detailed match events such as passes, shots, goals, fouls, and possession sequences, along with attributes such as timestamps, player identities, team information, and spatial coordinates.

The primary data source is the **StatsBomb** Open Data repository, which provides detailed event-level information from professional football matches, including passes, shots, fouls, and other in-game actions. Each event is associated with attributes such as match identifiers, team and player information, timestamps, and spatial coordinates, enabling fine-grained analysis of match dynamics (StatsBomb, 2023).

To complement this, event annotations from the **SoccerNet** dataset are incorporated. SoccerNet provides precisely labelled football events and, in some cases, associated contextual

descriptions, which are useful for both grounding and evaluation purposes (Giancola et al., 2018).

6.4.2. Textual Data

In addition to structured data, the system incorporates textual sources such as match reports, commentary, and summaries obtained from publicly available platforms, including sports news websites and encyclopaedic match pages.

These sources provide contextual and narrative descriptions of match events, which enhance the system's ability to generate coherent and human-readable outputs. Furthermore, they serve as a reference for qualitative evaluation by enabling comparison between system-generated summaries and human-written reports.

6.4.3. Metadata

Metadata plays an important role in improving retrieval accuracy and contextual relevance. This includes information such as team names, player profiles, match context (e.g., tournament stage), and temporal attributes.

Metadata is used during retrieval to filter and refine search results, thereby improving precision and reducing irrelevant information in the retrieved context.

6.4.4. Data Preprocessing

To enable compatibility with language models, structured event data is transformed into textual representations. Each event is converted into a natural language description while preserving key contextual attributes such as event type, time, location, and participating players. For example, an may be represented as "Player A completed a forward pass to Player B in the final third at minute 65."

These textual representations are then segmented into semantically meaningful chunks and enriched with metadata. The processed data is converted into vector embeddings and stored in a vector database to support efficient semantic retrieval.

In addition, standard preprocessing steps such as data cleaning, normalization, and formatting are applied to ensure consistency and usability across all data sources.

6.4.5. Justification of Data Selection

The decision to rely on structured and textual data, rather than raw video or audio inputs, is driven by both practical and methodological considerations. Structured datasets capture essential match dynamics and tactical patterns while significantly reducing computational complexity.

This approach ensures that the system remains **resource-efficient, scalable, and reproducible**, while still enabling the generation of contextually grounded and analytically meaningful outputs. It also aligns with the research objective of developing a system that balances analytical capability with real-world feasibility.

6.5. Data ingestion and metadata schema

Match data is organised by competition and season, with each match represented as a JSON file containing detailed event information. From these files, a standardised schema is extracted to support indexing and retrieval.

The key fields used in the schema are summarised in Table 6.3.

Table 4: Event Data Schema

Field	Description
match_id	Unique match identifier
team_id	Team associated with the event
player_id	Player performing the action
minute	Match minute (integer)
second	Match second (if available)
period	Match half (1 or 2)
x, y	Pitch location coordinates (0–120 by 0–80)
event_type	Type of event (e.g., Pass, Shot, Foul)
possession_id	Identifier for possession sequence

These attributes are used for both indexing and query filtering. For example, if a query references a specific team, time period, or match phase, the system can restrict retrieval using the corresponding metadata fields, improving both efficiency and relevance.

The StatsBomb Open Data format provides detailed event-level information, including spatial coordinates and possession identifiers, which enables structured representation of match dynamics and supports fine-grained retrieval.

6.6. Chunking strategy

Raw event data is transformed into textual "chunks" at multiple levels of granularity to balance detail and contextual understanding. This multi-level chunking strategy enables the system to support both fine-grained queries and higher-level tactical analysis.

Three levels of chunking are implemented. At the most granular level, event-level chunks are generated by converting each individual event into a single sentence. For example, *"23' – Team A – Player P took a shot from (82.4, 31.0); outcome: goal."* This representation enables precise retrieval of specific match actions.

At an intermediate level, time-window chunks aggregate events occurring within short temporal intervals (e.g., 30–60 seconds) into concise narratives. For instance, *"From 22'–23': Player X made a through ball to Player Y; shortly after, Team A had a corner kick."* This approach captures local context and short sequences of play.

At a higher level, sequence-based chunks are created by grouping events using possession identifiers. These chunks form coherent descriptions of attacking or defensive sequences, enabling analysis of tactical patterns such as build-up play leading to a goal.

All chunks are generated using fixed templates to ensure consistency and reduce variability in representation. A typical event-level template is structured as follows:

"<minute>': [<EventType>] <Team> – <Player> <description> at (x, y); outcome: <detail>."

The resulting textual chunks form the knowledge base for retrieval in the RAG pipeline. Each chunk retains links to its original metadata (e.g., match, team, possession), enabling efficient filtering and context-aware retrieval.

This hierarchical chunking approach improves retrieval flexibility by supporting both fine-grained evidence extraction and broader contextual understanding.

6.7. Embeddings and Vector DB setup

To enable semantic retrieval of relevant match events, textual chunks are converted into dense vector representations using pre-trained embedding models. These embeddings capture the semantic meaning of the text, allowing the system to retrieve contextually relevant information based on user queries.

Different families of embedding models are considered to balance performance and computational cost. Sentence-BERT (SBERT) models provide efficient and well-established semantic similarity capabilities, making them suitable for lightweight implementations. More recent models such as E5 are specifically trained for query–passage retrieval tasks and generally offer improved retrieval performance, although larger variants require higher computational resources. In addition, API-based embedding models (e.g., OpenAI embedding models) can provide high-quality representations but introduce external dependencies and usage costs.

Given the constraints of this project, priority is given to open-source embedding models that can be executed locally, such as SBERT or E5-base variants. This ensures cost efficiency while maintaining acceptable retrieval performance.

The generated embeddings are stored in a vector database to enable efficient similarity-based retrieval. Two primary approaches are considered. FAISS, a high-performance vector similarity library, is used for efficient indexing and retrieval of embeddings. For the scale of this study, a flat index (e.g., inner product or cosine similarity) is sufficient, while more advanced indexing techniques can be considered for larger datasets.

Alternatively, ChromaDB is explored as an open-source vector database that provides built-in support for metadata filtering and persistent storage. This simplifies implementation by allowing queries to be filtered directly based on attributes such as match identifiers or team information.

Metadata filtering plays a critical role in improving retrieval relevance. Since FAISS does not natively support metadata-based filtering, filtering is applied after retrieval using structured attributes such as match_id, team_id, or temporal constraints. In contrast, databases such as ChromaDB allow filtering to be applied during the retrieval process, improving efficiency and precision.

Overall, this embedding and vector database setup enables efficient and scalable retrieval of semantically relevant match events, forming the foundation for the retrieval component of the proposed Agentic RAG system.

6.8. Retrieval Strategy

This research adopts a hybrid retrieval-augmented generation (RAG) strategy. Conventional RAG primarily relies on semantic retrieval over embedded textual chunks, while knowledge graph RAG supports richer relational reasoning through explicitly modelled entity connections. In contrast, the proposed framework uses a practical hybrid approach that combines dense semantic retrieval with structured metadata filtering and optional keyword-based search. This design is intended to improve contextual grounding and retrieval relevance while remaining more lightweight and implementable than a fully graph-dependent architecture.

The Retrieval Agent is responsible for identifying the most relevant textual chunks from the knowledge base based on the user's query. The retrieval process combines semantic similarity search with structured metadata filtering to ensure both relevance and contextual accuracy. Initially, dense retrieval is performed by generating an embedding for the user query using the selected embedding model. This query embedding is compared with stored embeddings in the vector database to retrieve the top-N most similar chunks. This enables the system to capture semantic relationships beyond exact keyword matching.

To further improve precision, metadata-based filtering is applied using structured attributes such as match identifiers, team information, and temporal constraints (e.g., specific minutes or match periods). This ensures that retrieved results are aligned with the contextual requirements of the query and reduces irrelevant matches. A de-duplication step is then applied to remove overlapping or highly similar chunks, particularly in cases where time-window or sequence-level chunks may introduce redundancy. This helps maintain diversity and relevance in the retrieved evidence.

For queries containing specific keywords, such as player names or rare terms, an optional keyword-enhanced retrieval step can be employed. This combines dense retrieval with keyword-based search techniques (e.g., BM25), allowing the system to leverage both semantic similarity and exact matching. An optional re-ranking step may also be applied to refine the final set of retrieved chunks. This involves re-evaluating the top candidates using more precise but computationally expensive models, such as cross-encoders or lightweight language model-based ranking. Due to computational constraints, this step is selectively applied only when necessary.

Overall, this multi-stage retrieval strategy ensures that the selected evidence is both semantically relevant and contextually accurate, forming a reliable foundation for generating grounded summaries and tactical insights.

6.9. Retrieval Metrics

The performance of the retrieval component is evaluated using standard information retrieval metrics. Precision@K measures the proportion of relevant chunks among the top K retrieved results, where relevance is determined through manual assessment. Recall@K measures the proportion of all relevant chunks that are successfully retrieved within the top K results, based on pooled relevance judgments.

High Precision@K indicates that the retrieved evidence is largely accurate and useful, while high Recall@K reflects the system's ability to capture sufficient contextual information for downstream generation tasks.

6.10. Response Generation and Prompting

The Response Generation Agent is responsible for producing the final output by leveraging a prompt-based large language model. The agent integrates the retrieved textual chunks as contextual evidence and generates structured outputs such as match summaries, key moments, and tactical insights.

To ensure consistency and reduce the likelihood of unsupported outputs, structured prompt templates are used to guide the generation process. These templates explicitly separate the user query, retrieved evidence, and task instructions, ensuring that the model operates within a clearly defined context.

A typical prompt structure for a match summary task is presented below:

System: You are an expert football analyst. Use only the provided evidence to generate the response and avoid unsupported information.

User request: *{user_query}*

Contextual Evidence:

[{chunk_id_1}] {chunk_text_1}

[{chunk_id_2}] {chunk_text_2}

Task:

- Generate a concise summary of the match (150–200 words).
- Identify key moments with timestamps.
- Provide relevant tactical insights.
- If the required information is not available in the evidence, respond with *"Not enough information."*

This structured prompting approach ensures that the generated outputs remain grounded in retrieved evidence while maintaining clarity and coherence. By explicitly separating the query, evidence, and task instructions, the system improves interpretability and reduces the likelihood of hallucinated responses.

Additionally, referencing chunk identifiers in the prompt enables traceability of the generated output, allowing alignment between the response and supporting evidence. This enhances transparency and supports validation of the system's outputs.

6.11. Evaluation Plan

The proposed Agentic RAG system will be evaluated using a combination of **quantitative metrics and qualitative assessment**, ensuring a comprehensive evaluation of both retrieval performance and generation quality. The evaluation follows a **controlled experimental setup**, where the proposed system is compared against a **naive RAG baseline** under identical conditions.

6.11.1. Evaluation Setup

The evaluation is conducted using a controlled experimental design, where both the Agentic RAG system and the naive RAG baseline are tested on the same set of input queries. These

queries are designed to reflect realistic user scenarios, including match summaries, player performance analysis, and tactical insights.

A fixed evaluation dataset will be constructed using approximately **15-20 football matches** selected from publicly available structured event datasets such as StatsBomb and SoccerNet. These matches will be chosen to represent a range of game contexts, including different teams, match dynamics, and event distributions. From this dataset, approximately **40-60 evaluation queries** will be created, covering multiple query types such as overall match summarisation, key moments, player-specific analysis, and tactical interpretation.

Using the same set of matches and queries for both systems ensures a fair and controlled comparison under identical experimental conditions.

6.11.2. Retrieval Evaluation

The effectiveness of the retrieval component is evaluated using standard information retrieval metrics:

- **Precision@K (P@K):** Measures the proportion of relevant documents among the top-K retrieved results
- **Recall@K (R@K):** Measures the proportion of relevant documents successfully retrieved

Relevance is determined based on whether the retrieved events or documents correspond to the ground-truth match context associated with the query. This is established using labelled event data and known match outcomes.

6.11.3. Generation Evaluation

The quality of generated summaries and insights is evaluated using established natural language generation metrics:

- **ROUGE:** Measures overlap with reference summaries (recall-oriented)
- **BLEU:** Measures precision of generated text against references
- **BERTScore:** Evaluates semantic similarity between generated and reference text

Reference summaries are constructed using human-written match reports and curated textual descriptions from publicly available sources.

6.11.4. Human Evaluation

In addition to automatic metrics, qualitative evaluation will be performed using human judgment to assess aspects that cannot be fully captured by numerical metrics.

The human evaluation will involve approximately 3-5 evaluators with relevant familiarity in football analysis, sports media, or data-driven interpretation of match events. Evaluators will assess a selected subset of generated outputs using the following criteria:

- **Groundedness:** Whether the output is factually consistent with retrieved data
- **Tactical Usefulness**: Whether the insights provide meaningful analytical value
- **Coherence:** Whether the output is logically structured and readable

Each criterion will be rated on a 5-point Likert scale, and inter-rater reliability will be measured using Krippendorff's alpha to ensure consistency across evaluators.

6.11.5. Baseline Comparison

The proposed Agentic RAG system is compared against a naive RAG baseline, which consists of a single-stage retrieval and generation pipeline with fixed chunking and no explicit reasoning or metadata filtering.

The comparison evaluates improvements in:

- Retrieval relevance
- Summary quality
- Depth of tactical insights

This enables a clear assessment of the benefits introduced by the agent-based architecture.

6.11.6. Ablation Study

To further validate the contribution of individual components within the proposed Agentic RAG framework, an ablation study will be conducted. This analysis aims to assess the impact of each architectural component by systematically removing or modifying them and observing the resulting changes in system performance.

The study begins with the **full Agentic RAG model**, which includes metadata-aware retrieval, a reasoning agent, and an evaluator agent, serving as the reference configuration. To understand the importance of contextual retrieval, a variant without metadata filtering will be evaluated, where retrieval is performed without constraints such as event type, player, or temporal context.

To assess the role of agentic reasoning, the reasoning agent will be removed, and outputs will be generated directly from retrieved context without multi-step analysis. Similarly, the evaluator agent will be excluded in another configuration to measure the impact of output validation and refinement on overall quality.

In addition, the effectiveness of data representation will be analysed by replacing hierarchical chunking with a flat, event-level chunking strategy. Finally, a standard naïve RAG pipeline without agentic components will be used as a baseline for comparison.

All configurations will be evaluated using the same dataset, queries, and metrics defined in the previous sections to ensure consistency and fairness. This ablation study will provide a detailed understanding of how each component contributes to retrieval performance, reasoning capability, and the quality of generated tactical insights.

6.11.7. Success Criteria

The proposed system will be considered successful if it demonstrates measurable improvements over the baseline across multiple evaluation dimensions. Specifically:

- Improvement in **retrieval metrics** (Precision@K, Recall@K)
- Higher scores in **generation metrics** (ROUGE, BLEU, BERTScore)
- Improved **human evaluation scores**, particularly in tactical usefulness and groundedness

Additionally, consistent performance across multiple query types will indicate robustness of the system.

6.11.8. Error Analysis

To further strengthen the evaluation, an error analysis will be conducted to identify limitations and failure cases of the system. This includes:

- Cases of incorrect or irrelevant retrieval
- Instances of hallucinated or unsupported outputs
- Scenarios where tactical insights are incomplete or misleading

This analysis will provide insights into areas for improvement and help validate the effectiveness of the proposed architecture.

6.12. Implementation Details

The proposed system is implemented using a Python-based technology stack to support efficient development and integration of machine learning components. Python is selected due to its extensive ecosystem for natural language processing, machine learning, and data handling.

Key libraries include transformer-based frameworks for embedding generation, along with vector similarity libraries and database tools for indexing and retrieval. The retrieval component is implemented using vector search frameworks such as FAISS or lightweight vector databases that support metadata filtering and persistence.

For response generation, the system utilises a large language model accessed either through an API-based service or an open-source model deployed locally. This provides flexibility in balancing performance, cost, and computational requirements.

The development environment consists of cloud-based notebooks or local virtual machines, enabling experimentation and scalability without requiring dedicated infrastructure. This setup supports rapid prototyping while remaining compatible with resource constraints.

6.13. Budget and Resource Considerations

The proposed methodology is designed to operate within modest computational and financial constraints by avoiding resource-intensive multimodal processing.

Embedding generation is performed as a one-time operation and can be executed on standard CPU environments or accelerated using cloud-based GPUs where available. Vector indexing and storage require minimal computational overhead for the dataset scale considered in this study.

For response generation, the use of API-based language models introduces a usage-based cost that depends on the number of tokens processed. However, given the limited scale of evaluation (e.g., a small number of matches and queries), the overall cost remains low and manageable within a typical student project budget. Alternatively, open-source models may be used to eliminate API costs, with the trade-off of higher local computational requirements.

Several cost optimisation strategies are applied. Embeddings are cached to avoid recomputation, reducing repeated processing overhead. Metadata filtering is applied prior to generation to limit the amount of context passed to the language model, thereby reducing token usage. In addition, the number of retrieved chunks is kept minimal while maintaining sufficient coverage for accurate responses. Smaller models may also be used during development to further reduce cost.

6.14. Ethical Considerations

The study adheres to ethical standards in data usage, system design, and evaluation. All datasets used in this work are publicly available and intended for research purposes. Appropriate attribution is provided for data sources, and no sensitive or personal data is processed beyond publicly available information such as player and team names.

To address the risk of incorrect or misleading outputs, the system is designed to generate responses grounded in retrieved evidence. When sufficient evidence is not available, the system explicitly indicates uncertainty rather than producing unsupported claims.

Bias is also considered during evaluation, particularly in the interpretation of match events and tactical insights. Human evaluators are instructed to assess outputs objectively, and multiple evaluators are used to reduce subjectivity.

Transparency is ensured by linking generated outputs to the underlying evidence used during retrieval. This allows verification of claims and improves trust in the system.

Finally, the limitations of the approach are acknowledged. The absence of video or multimodal data restricts the system's ability to capture certain aspects of gameplay, such as off-ball movements or positional dynamics. These limitations are recognised and discussed as part of the evaluation.

7. Requirements and Resources

This chapter outlines the data, software, hardware, and computational resources required for the design, implementation, and evaluation of the proposed Retrieval-Augmented Generation (RAG)-based framework for automated football match summarisation and tactical insight generation. The selection of resources is guided by the objective of developing a scalable, cost-efficient, and reproducible system within the constraints of an academic research environment.

7.1. Data Requirements

The proposed system relies on publicly available structured football datasets to support both development and evaluation.

The primary data source is the StatsBomb Open Data repository, which provides detailed event-level information such as passes, shots, fouls, player actions, timestamps, and spatial coordinates. This dataset enables fine-grained representation of match dynamics and forms the core input for the retrieval pipeline (StatsBomb, 2023).

In addition, the SoccerNet dataset is used to complement event annotations and support evaluation tasks. It provides labelled football events and, in some cases, associated textual descriptions that can be used for validation and grounding (Giancola et al., 2018).

For qualitative evaluation, human-written match summaries from publicly available sources are utilised as reference material. These summaries provide contextual narratives that serve as a benchmark for comparing system-generated outputs.

7.2. Software Requirements

The system is implemented using a modern Python-based technology stack that supports efficient development, dependency management, and integration of machine learning components.

Python3 is used as the primary programming language due to its extensive ecosystem for natural language processing and machine learning. Dependency and environment management are handled using **UV**, enabling fast, reproducible, and lightweight setup of project environments.

The core system pipeline is developed using **LangChain**, which facilitates orchestration of the Retrieval-Augmented Generation workflow, including query handling, retrieval, and response generation. Embedding models are implemented using **Hugging Face Transformers** and **Sentence Transformers**, while vector storage and retrieval are supported using **FAISS** or **ChromaDB**.

The development environment is based on **Visual Studio Code (VS Code)**, which provides an integrated platform for coding, debugging, and project management. Version control is managed using **Git**-based repositories to ensure reproducibility and systematic tracking of changes.

This software stack enables modular implementation of the Agentic RAG architecture while supporting flexibility, scalability, and ease of experimentation.

7.3. Hardware Requirements

The proposed system is designed to operate within modest computational constraints and does not require specialised high-performance infrastructure.

A standard computing environment with a multi-core CPU is sufficient for data preprocessing, retrieval operations, and system integration. For embedding generation, optional access to a GPU (e.g., via cloud-based platforms such as Google Colab) can accelerate processing, although it is not mandatory.

Memory and storage requirements are minimal, as the dataset size and vector index scale are manageable within typical academic computing environments. The absence of multimodal processing significantly reduces computational overhead.

7.4. Computational Resources and Budget

The computational cost of the system is minimised through efficient design choices and lightweight architecture.

Embedding generation is performed as a one-time operation and cached for reuse, reducing repeated computational overhead. Vector indexing and retrieval operations are efficient and can be executed on standard hardware without significant resource consumption.

For response generation, the system may utilise API-based large language models or open-source alternatives. When using API-based models, the cost is determined by token usage. However, given the limited scope of evaluation (e.g., a small number of matches and queries), the overall cost remains low and manageable within a typical student project budget.

Cost optimisation strategies include limiting the number of retrieved chunks, applying metadata filtering prior to generation, and using smaller models during development phases. Open-source models may also be used to eliminate API costs, with a trade-off in computational requirements.

7.5. Resource Constraints and Design Considerations

This research is conducted under practical constraints, including limited computational resources and absence of specialised infrastructure. As a result, the system intentionally avoids computationally intensive multimodal processing involving video and audio data.

Instead, the proposed framework focuses on structured event data and efficient retrieval mechanisms, ensuring that the system remains scalable, reproducible, and suitable for academic and real-world deployment scenarios.

These constraints inform key design decisions, emphasising efficiency, modularity, and cost-effectiveness, while maintaining the ability to generate accurate and context-aware outputs.

8. Project Timeline and Gantt Chart

This chapter outlines the planned timeline for the execution of the proposed research. The project is structured into multiple phases, covering data preparation, system design, implementation, evaluation, and documentation. The timeline follows a week-based structure to provide flexibility while ensuring systematic progression of the research.

The total duration of the project is estimated to be **twelve weeks**, allowing sufficient time for development, experimentation, evaluation, and final documentation.

8.1. Project Timeline

The key phases and corresponding activities are summarised in Table 8.1.

Table 5: Project Timeline

Phase	Activities	Duration
Phase 0	Topic selection, Data collection and gathering, Research planning	Week 0–1
Phase 1	Data preparation, preprocessing, and validation	Week 1–2
Phase 2	System design and architecture development	Week 2-4
Phase 3	Implementation of embedding and retrieval components	Week 4-6
Phase 4	Development of response generation and prompting	Week 6-8
Phase 5	Evaluation using retrieval and generation metrics	Week 8–10
Phase 6	System refinement and optimisation	Week 10–11
Phase 7	Documentation and final report preparation	Week 11–12

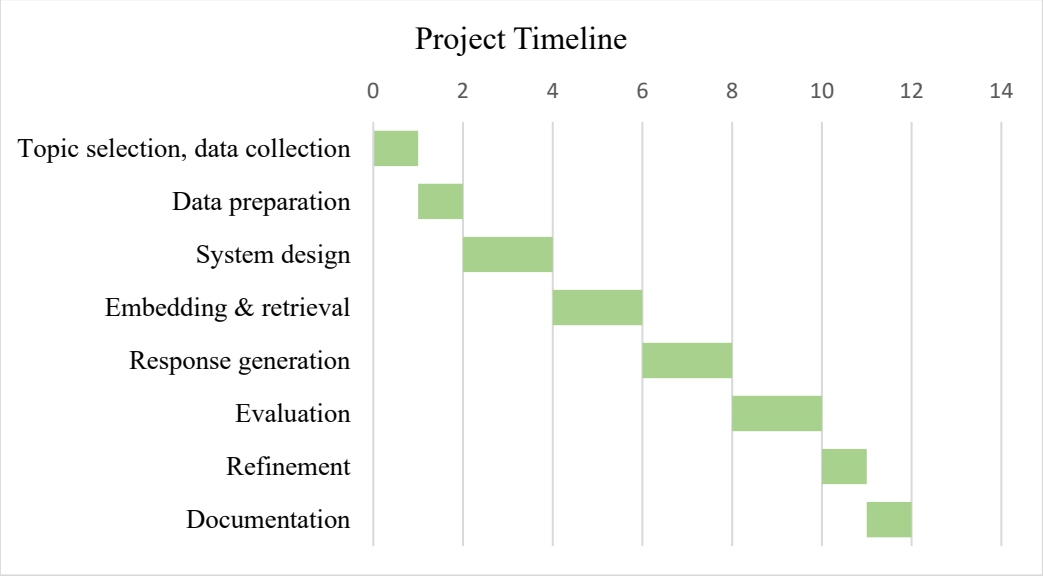


Figure 2: Project Timeline

8.2. Gantt Chart Representation

The timeline can be represented as a Gantt chart to illustrate the progression and overlap of activities.

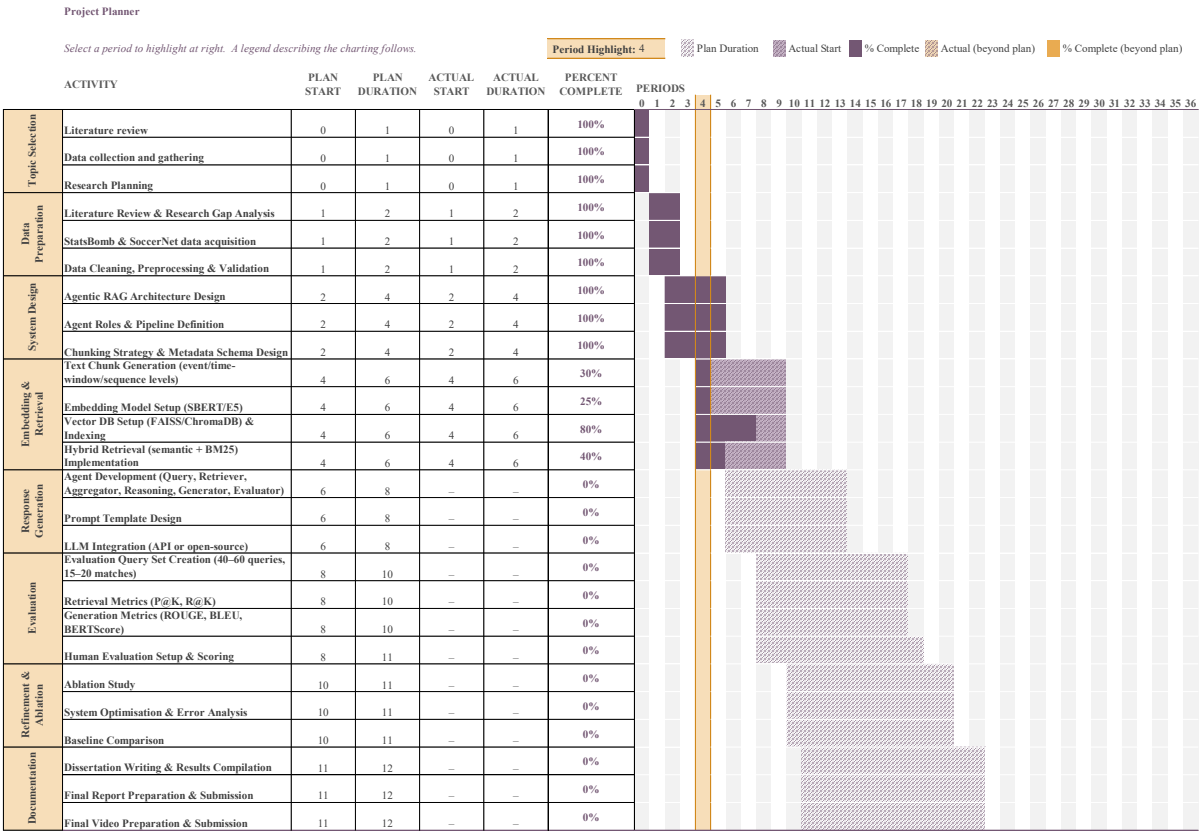


Figure 2: Gnatt Chart

8.3. Timeline Justification

The timeline is designed to ensure a balanced distribution of effort across all stages of the research. Initial weeks focus on preparing the dataset and establishing the system design. The middle phase emphasises implementation of the Retrieval-Augmented Generation pipeline, including retrieval and generation components.

Evaluation and refinement are conducted in later stages to improve system performance based on quantitative and qualitative metrics. The final phase is dedicated to documentation and consolidation of findings, ensuring a complete and well-structured research submission.

This phased and overlapping approach ensures efficient utilisation of time while maintaining flexibility for iterative improvements during development.

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